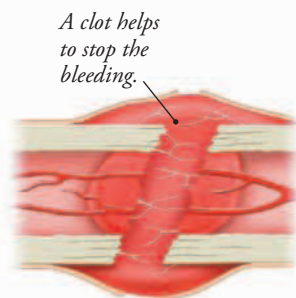
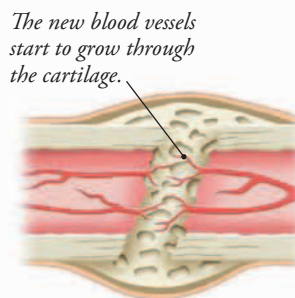


HEALING BONES

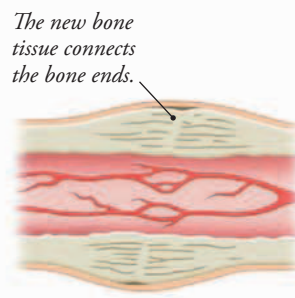
If bones break or fracture, the healing process begins immediately. Within hours, as you can see in this section of a broken long bone, a blood clot forms to stop bleeding from vessels inside the bone. Then rebuilding begins. Cartilage is used to make a temporary repair and is gradually replaced by stronger bone cells. The whole process takes about 12 weeks, but is longer for leg bones that carry the body's weight.



▲ **THREE HOURS**
Within hours a blood clot forms between the bone ends.

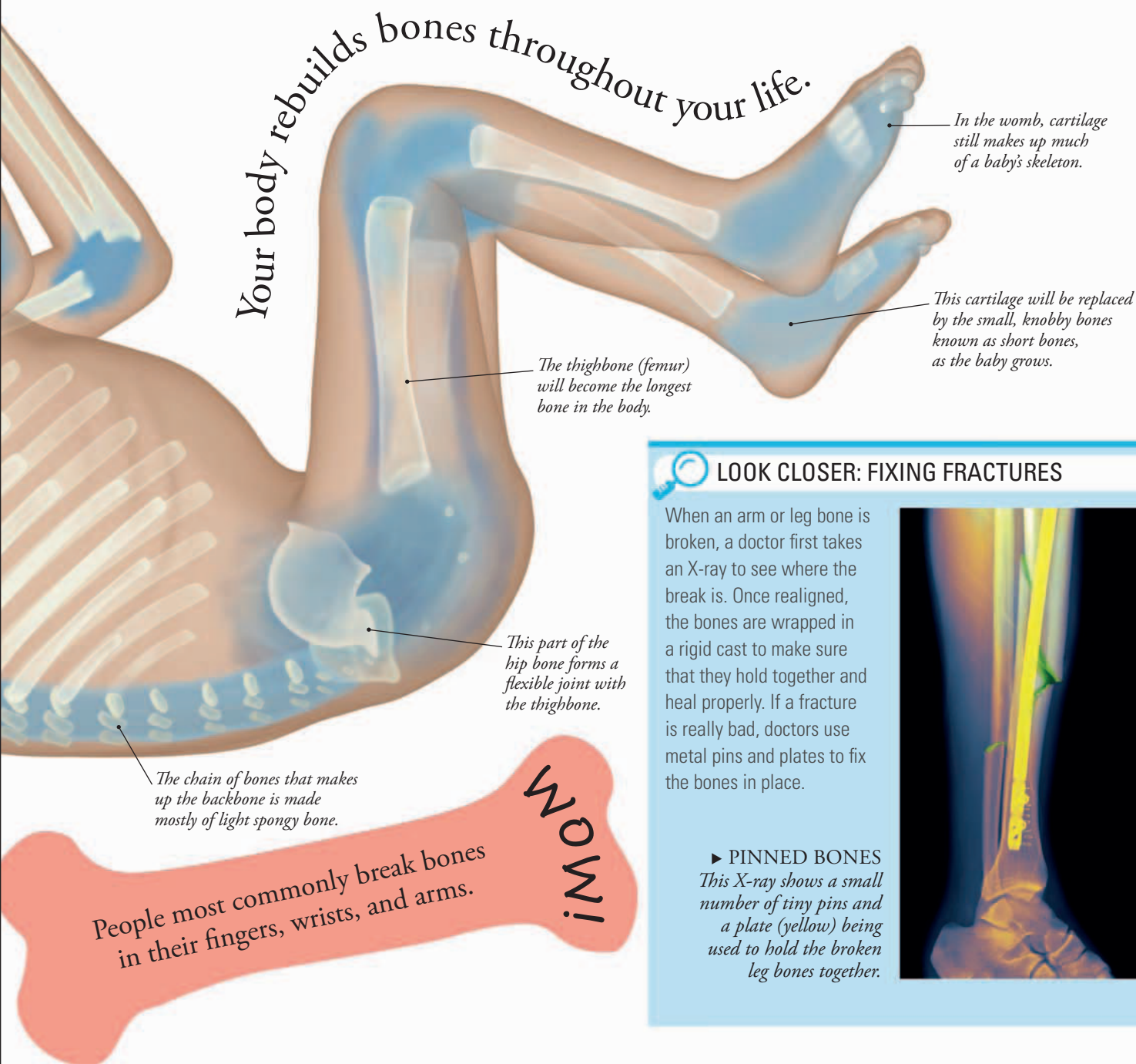


▲ **THREE WEEKS**
The cartilage replaces the clot to bridge the gap in the bone.



▲ **THREE MONTHS**
The bone replaces the cartilage and the break is repaired.

Your body rebuilds bones throughout your life.



In the womb, cartilage still makes up much of a baby's skeleton.

This cartilage will be replaced by the small, knobby bones known as short bones, as the baby grows.

The thighbone (femur) will become the longest bone in the body.

This part of the hip bone forms a flexible joint with the thighbone.

The chain of bones that makes up the backbone is made mostly of light spongy bone.

People most commonly break bones in their fingers, wrists, and arms.

WOW!

LOOK CLOSER: FIXING FRACTURES

When an arm or leg bone is broken, a doctor first takes an X-ray to see where the break is. Once realigned, the bones are wrapped in a rigid cast to make sure that they hold together and heal properly. If a fracture is really bad, doctors use metal pins and plates to fix the bones in place.

► **PINNED BONES**
This X-ray shows a small number of tiny pins and a plate (yellow) being used to hold the broken leg bones together.

